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Seed Availability and Small Mammal Populations: Insights from Mediterranean Forests

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Abstract: Plant–animal interactions play a crucial role in ecosystem functioning, especially through seed dispersal mechanisms. Understanding how small mammal populations respond to seed availability is essential for ecosystem management and biodiversity conservation, especially in the context of habitat loss and climate change. We conducted a 10-year study in mixed Mediterranean oak–beech forests to investigate the population dynamics of common small mammal species in response to seed availability. Our findings revealed distinct responses among species, influenced by life history traits, foraging behaviour, and diet. Wood mice (*Apodemus sylvaticus*) showed a rapid population increase with seed availability both in the same year of seed fall and the following year, suggesting a flexible foraging strategy and a dependence on arboreal seed producers. Yellow-necked mice (*Apodemus flavicollis*) revealed immediate population increases in response to seed availability in the autumn, probably because of their arboreal habits and preference for exploiting seeds prior to maturation. Bank voles (*Clethrionomys glareolus*) showed responses with population peaks in years following high seed availability, indicating a slower demographic response to resource abundance. Surprisingly, the greater white-toothed shrew (*Crocidura russula*) responded indirectly to seed availability in Mediterranean forests, suggesting complex interactions with seed-associated invertebrates or dependence on other variables not considered. Our findings highlight the importance of understanding how changes in seed availability influence the population ecology of small mammals, with significant implications for the conservation and management of Mediterranean forest ecosystems in the context of climate change and recurrent droughts. These results emphasise the need to consider species interactions, resource availability, and climate change in the conservation and management of evolving ecosystems.

Keywords: Mediterranean mixed forests; plant–small mammal interactions; rodents; seed availability; shrews; seed predators

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1. Introduction

Plant–animal interactions play important ecological functions and are a fundamental aspect of ecosystem structure. Efficient seed dispersal is crucial for plant population stability, maintaining functional connections among metapopulations and colonising new habitats [1,2]. With the contemporary increase in habitat loss and fragmentation, dispersal has become even more relevant, especially in forests, as it can influence species composition and ecosystem functioning at regional and global scales [3–5]. On the other hand, the relationship between seed production by plant species and the population response of small mammals is particularly important [6–9].

Small mammals, such as small rodents, can be considered seed dispersers and predators, playing a crucial role in seed dispersal and vegetation regeneration and thus influencing ecosystem dynamics and diversity [10]. This group is generally more tolerant to habitat fragmentation and less prone to extinction than large herbivores [11,12]. They are

distributed on all continents with high species richness and high population densities [13] and are often considered keystone species that affect many ecosystem processes [14]. Seed production by plant species is a key factor in the ecology of small mammals, providing an abundant and nutritious food source that can influence their reproduction, survival, and movement within the ecosystem [15–17].

In temperate and boreal forests, massive seed production can trigger a significant increase in small mammal abundance [18,19]. This phenomenon occurs when a plant species locally synchronises its seed production in the same year, creating an overabundance of food that can be exploited by small mammals, whose reproduction is closely correlated with food availability [19–21]. This reproductive strategy has long been discussed in the literature, and it is not perfectly understood for most species, although it seems particularly clear in wind-pollinated trees of the northern hemisphere such as beech and oak trees [22–26]. Therefore, the relationship between seed production by plant species and small mammal population responses is crucial for forest ecosystem functioning [27]. Understanding this relationship is essential for the conservation and management of biodiversity in protected areas, as well as for the restoration of degraded ecosystems affected by climate change [28].

Ecologists have long been interested in understanding how resource pulses elicit different consumer responses and how these responses affect community dynamics through competitive interactions [29]. Many animal species that feed on trees seeds or fruits increase in numbers in response to an increased food supply. For example, *Apodemus* species, which store acorns during the autumn season, reach peak levels the following year because of increased winter survival. This phenomenon has been observed in the forests of the northern hemisphere [30–32].

The relationship between seed production and the abundance and diversity of small mammals has been studied in diverse ecological contexts [33–39]. However, a better understanding of these processes is required in Mediterranean ecosystems, where global change models predict extreme decreases in precipitation and increases in temperature, directly affecting forest dynamics [40,41]. This will increase the frequency and intensity of dry spells characteristic of this climate, which in turn will increase the soil aridity. Thus, Mediterranean forests are likely to suffer warming-related reductions in growth and production [40,42,43], and the Mediterranean domain is projected to shift towards the North in the coming decades [44].

Seed production in Mediterranean ecosystems forest is influenced by key environmental factors, including climatic variability, water availability, and interactions with other organisms [45,46]. Some of the most abundant and relevant seed producers in the northeast of the Iberian Peninsula are different species of oak (*Quercus petraea*, *Quercus ilex*, and *Quercus canariensis*) and beech (*Fagus sylvatica*). Previous studies have identified several climatic factors that affect the synchronous seed production of these species. For example, in the case of beech, high seed production depends on the summer temperature in the two years prior to the sampling year [45,47,48]. On the other hand, the acorns of the different *Quercus* species mature during the summer, when water limitation and elevated temperatures usually generate high abortion rates in tree species [41,46,49]. The same is true for beech, whose growth and production are limited by the hot and dry summers of the Mediterranean climate [43,49,50]. The prediction of climatic extremes resulting from global warming is expected to threaten the functioning of Mediterranean ecosystem forests, where summer droughts and forest fires are recurrent events, which may have a significant impact on seed production and ultimately on small mammal populations [51,52].

It has been shown that some common small mammal species in Mediterranean forests, such as wood mice (*Apodemus sylvaticus*), yellow-necked mice (*Apodemus flavicollis*), and bank vole (*Clethrionomys glareolus*) [53], respond to seed availability [31,37,54]. Furthermore, in addition to food availability, other factors such as intraspecific and interspecific competition, predation, and vegetation structure can also affect small mammal populations in Mediterranean forests [55]. The interplay among these factors can create spatial and

temporal patterns in the distribution and abundance of small mammal species in response to seed production [29,56,57].

Knowledge of how seed availability influences the population dynamics of small mammals can provide valuable information for biodiversity monitoring and management of the ecosystems in protected areas such as those found in the northeast of the Iberian Peninsula. Moreover, understanding the mechanisms regulating the relationship between seed production and small mammals may have implications for restoring degraded ecosystems or those affected by climate change. Therefore, restoring the dynamics of plant–animal interactions, including seed dispersal by small mammals, may be essential for the recovery of vegetation and ecosystem functionality in forests [58,59].

The main objective of this study was to investigate the population dynamics of the most common small mammal species in response to seed availability in mixed oak (*Q. petraea*, *Q. canariensis*, and *Q. ilex*) and beech (*F. sylvatica*) forests in a Mediterranean area. This region encompasses a diverse range of climates and forest landscapes, creating a gradient that includes varying environmental conditions. The small mammal communities were monitored over 10 years (2012–2022) in seven live trapping stations sampled in the spring and autumn, and the availability of seeds was measured at representative habitats. This transition along the gradient represents a unique opportunity to understand how vegetation changes and ecosystems respond to spatial variations, which is crucial for understanding and conserving biodiversity and natural resources in diverse forest environments.

We expected that species responses to seed availability would be associated with life-history traits (e.g., the timing of reproduction, [37,60]), foraging behaviour (e.g., degree of arboreality [61,62]; seed-hoarding behaviour [36]), and feeding strategies (e.g., insectivores, granivores [33,63,64]). Therefore, we expected that tree-dwelling granivorous rodents would have faster population responses to seeds if they were available before seed fall and delayed responses associated with seed availability on the ground and seed caches. In addition, owing to their scatter-hoarding behaviour, we expected floor-dwelling rodent species—having limited access to seeds only when they fall from the trees—would have delayed population responses mostly caused by this behaviour. Also, we predicted that insectivorous shrews would have only delayed responses associated with the increase in seed predatory insects. Finally, we also expected that specific population responses to seed availability would affect community diversity estimates.

2. Methods

2.1. Study Area

The study area represented an extensive gradient of climates and forest landscapes ranging from low-elevation coastal environments under Mediterranean influence, towards elevated areas in the pre-coastal mountain range with marked Euro-Siberian influence [65]. We sampled seven habitats (Figure 1) that were considered representative of the two prevailing climatic and forest domains (i.e., Mediterranean and Euro-Siberian), where seed measurements were performed in mixed oak (*Q. petraea*/*canariensis*/*ilex*) and mixed beech (*Fagus*/*Quercus*) forest areas.

Our work in the pre-coastal range focused on the protected areas of Montseny Natural Park Biosphere Reserve and Guilleries-Savassona Natural Area. The mean annual temperature is 11.3 °C and the annual rainfall is 1415.2 mm [66]. Moreover, the highest peak, "Turó de l'Home", has an altitude of 1712 m [67]. The vegetation is mainly composed of *F. sylvatica*, *Q. petraea*, *Corylus avellana*, and *Acer opalus* mixed with *Q. ilex* and *Ilex aquifolium* [68], but our sites were mainly located in beech forests.

The study area of the coastal range is located in Montnegre and Corredor Natural Park. This area is smaller (approximately 150 km^2), and the culminating ridge of Montnegre is slightly more than 3 km long above 700 m in altitude [69]. The mean annual temperature is 14.5 °C and the annual rainfall is 1067.4 mm [70]. The vegetation is composed of *Q. petraea*, *Q. canariensis*, and *Q. ilex* [68].

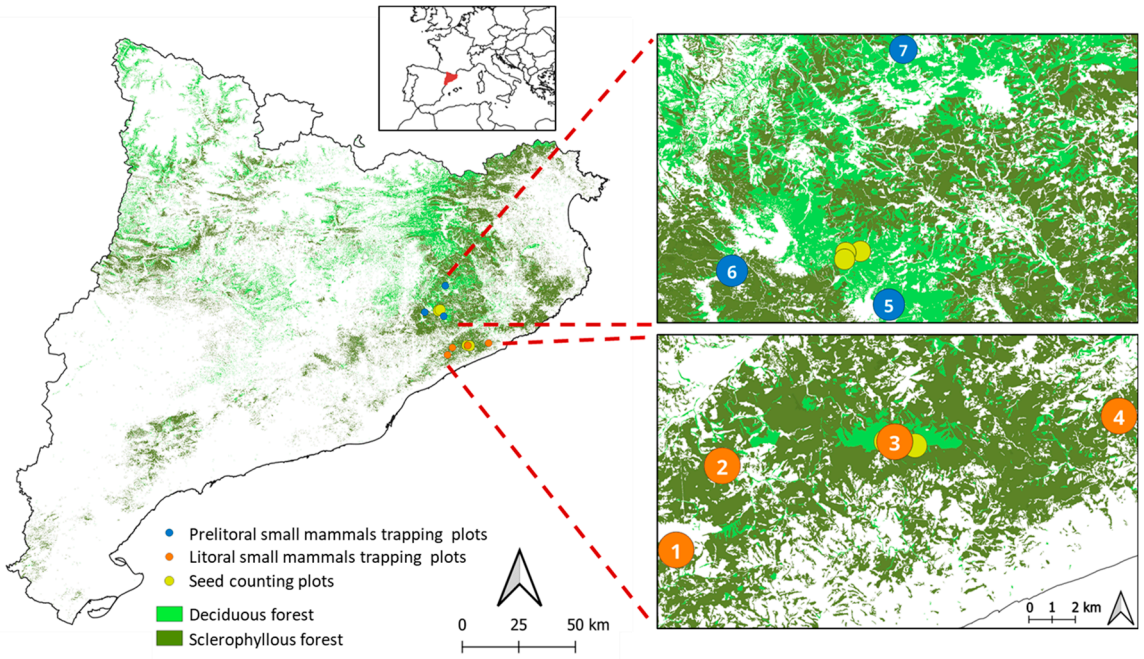


Figure 1. Location of this study in Catalonia, Spain. Small mammal sampling plots are represented in blue (*F. sylvatica* forest area) and orange (*Quercus* sp. forest area). Seed counting plots are represented in yellow. Deciduous and sclerophyllous forests are represented in light and dark green, respectively. Numbers correspond to the Plot ID, as in Figure 2 and Table 1. This figure was prepared with QGIS (Development Team, 2023), and geographical information from "Institut Cartogràfic i Geològic de Catalunya" was used.

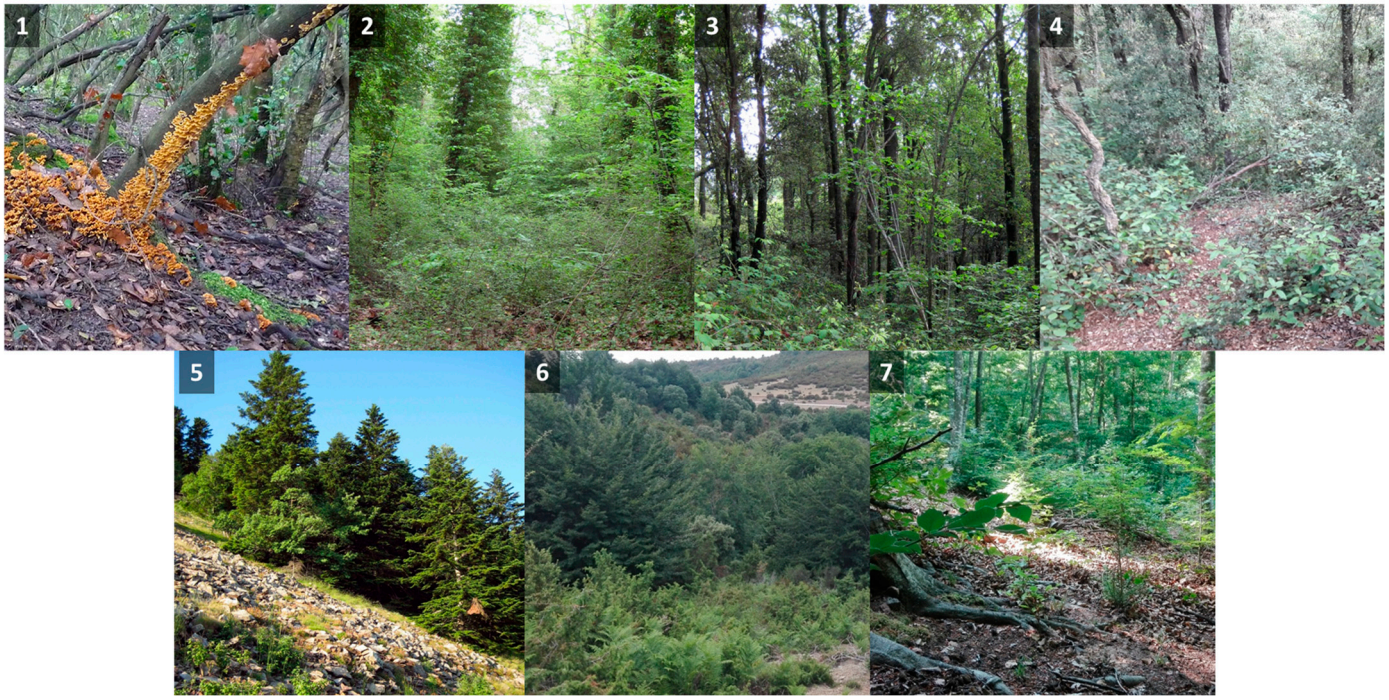


Figure 2. Small mammal sampling plots. Upper row: sampling plots from the coastal range. Lower row: sampling plots from the pre-coastal range. The numbers of each image correspond to the Plot ID in Figure 1 and Table 1 (photos from www.semice.org; accessed on 14 December 2023).

Table 1. Description of small mammal sampling plots. Plot_ID: number associated with the plot. Altitude: height at sea level expressed in meters. Scrub: percentage of shrub cover. Forest: Percentage of tree cover. Urban: percentage of urban area.

Plot_ID	Name	Vegetation	Altitude	Scrub	Forest	Urban
1	Alzinar Sot del Fangar	*Quercus*	516	3.62%	95.57%	0.82%
2	Riera de Vallgorguina	*Quercus*	193	10.32%	80.58%	9.10%
3	Roureda del Turó Gros	*Quercus*	750	0.00%	100.00%	0.00%
4	Turó de Miralles	*Quercus*	257	5.39%	94.61%	0.00%
5	Avetosa de Passavets	*Abies–Fagus*	1442	9.86%	90.14%	0.00%
6	Fageda de la Cortada	*Fagus*	1200	25.58%	74.42%	0.00%
7	Fageda de les Valls	*Fagus*	908	1.99%	98.01%	0.00%

2.2. Small Mammal Sampling

Small mammal abundance was assessed using a capture–mark–recapture study following the SEMICE (www.semice.org; accessed on 14 December 2023) monitoring scheme [71] in live trapping plots placed in representative habitats of the study area. The plots selected included the following main habitats (Table 1): riverbed (one plot), holm oak (*Q. ilex*, two plots), oak grove (*Q. canariensis*, one plot), beech (*F. sylvatica*, two plots), and fir (*A. alba*, one plot). Sampling was conducted twice a year, in spring and autumn, to cover the life cycle of the target species [72,73]. The time series spanned ten years, from spring 2012 to autumn 2022. The sampling plots were grids of 36 traps, using both a 6 × 6 and 9 × 4 (riverbed) scheme design. To mitigate size-related biases in small mammal community assessments, Longworth traps (Longworth Scientific Instrument Co., Oxford, UK; 25 × 6.5 × 8.5 cm [74,75]) and Sherman traps (Sherman folding small animal trap; 23 × 7.5 × 9 cm; Sherman Co., Tallahassee, FL, USA) were alternated in position one at a time [76]. The traps were placed 15 m apart, baited with a piece of apple to prevent dehydration and with a mixture of tuna and flour as an energy source. A handful of hydrophobic cotton was included as bedding material and thermal isolator [77,78]. The traps were set for three consecutive nights and checked in the early morning of the first, second, and third day. The species captured were identified, sexed, weighed, tagged with ear-tags in the case of rodents (Style 1005-1, National Band Co., Newport, RI, USA) and fur clips for shrews, and released at the point of capture [79]. The number of different individuals captured over the three days (excluding recaptures) was considered an index of relative abundance. Owing to the high mean values of detectability, indices of abundance of the species analysed were assumed to be unbiased (i.e., low probability of non-detecting the species when actually present) [53].

The seven sampling plots covered the range of habitats and climates observed in the coastal and the pre-coastal ranges of the Barcelona province (Figure 2). In the pre-coastal mountain range, the plots were located in the beech forest domain, while in the coastal mountain range, they were located in areas of mixed oak forests. Elevations of the plots ranged from 193 m to 1442 m above sea level.

2.3. Seed Availability Data

Seed availability was assessed from 2012 to 2021 in three plots situated in Mediterranean mixed forests (*Q. canariensis*, *Q. petraea*, *and Q. ilex*) and three plots in beech-dominated forests, by counting and weighting the seeds—intact and preyed/infested—in 20 small quadrats (50 × 50 cm, 0.25 m^2) fixed on the ground [37,40]. Each quadrat was set under a tree and placed 25–30 m apart forming a 5 × 4 grid. Seed counting was performed six times a year every two weeks, from the end of August to the end of November. To prevent double counting, the sampling quadrats were cleaned after each check. We considered the whole number of seeds counted during each check and plot as the seed index (20 quadrats on a 5 m^2 surface), calculating the mean annual count of the three plots as an estimate of seed availability in the nearby small mammal plots. Since the number of seeds

and their weight were highly correlated (r = 0.97), we considered only the first variable for the analyses.

To assess the variability in seed counts across spatial and temporal scales and to validate the use of annual averaged seed counts as a reliable indicator of seed availability in our study area, we conducted statistical analyses using Linear Mixed Models [80] in RStudio software [81]. Specifically, we employed LMMs where seed counts at each sampling plot served as the response variable, with year included as a fixed effect and plot as a random effect. This approach allowed us to compare the explanatory power of the fixed predictor (year) against the variability attributed to the random factor (plot). By employing these models, we aimed to evaluate whether seed counts varied more significantly among sampling plots or across different years, thereby ensuring robustness in our characterisation of annual seed availability dynamics.

2.4. Data Analysis

To assess the effects of seed availability on the relative abundance of small mammals we used Generalized Linear Mixed Models [82]. The number of individuals captured for each species during each trapping season was included as the response variable, and the predictor variables were the seed index of the previous year (to test for delayed population responses of small mammals [33–35,57,82]), the seed index of the corresponding year (to test the response of small mammals in the same year), and the sampling season (spring or autumn). We also included the two-level interactions between season and each seed count. Plot, year, and area (prelitoral or litoral) were included as random factors to account for the natural variability among locations and data collection times. Variance inflation factors were calculated to evaluate possible multicollinearity in the explanatory variables (VIF; R package car; [83]). Since seed availability showed synchrony (both spatially and temporally, [84–86]), we also expected similar responses to seed availability of synchronic small populations [87,88]. Since models with a Poisson distribution showed high values of overdispersion, all models were run with a negative binomial error distribution to cope with overdispersion [89] by means of the "log" link function, R package *lme4* [90].

Additionally, we used LMMs to analyse the role of seed availability on the whole community of small mammals (i.e., changes in species richness and evenness) using Hill's numbers [91], which allowed us to calculate the effective number of species (or equiprobable species) by using the parameter "q", which indicates the sensitivity of the index to the relative abundance of species. Thus, if q = 0 (Order 0), the result indicates species richness (number of species); if q = 1 (Order 1), it 1 yields the exponential of the Shannon index; if q = 2 (Order 2), the inverse Simpson index is obtained (Hsieh et al., 2016). The same structure as the GLMM was used, but in that case, different diversity order values were used as the response variable (0 = species richness; 1 = species diversity; 2 = species dominance), resulting in three different model sets [92]. To calculate the diversity value according to each order in each plot and each year, we used the R package *vegan* [93].

The GLMM and LMM were built using the R package *lme4* [90], using the *dredge* function of the package *MuMIn* [94], and model complexity (parsimony) was assessed by the corrected Akaike information criterion (AICc), selecting those with ΔAICc ≤ 2 with respect to the lowest-valued model [95]. For the selected models, we calculated pseudo-R^2 values by means of the R function *r.squaredGLMM* and the delta method for variance estimation [96].

3. Results

3.1. Seed Availability

The main seed-producing species at the sampling plots were beech (*Fagus sylvatica*) and *Quercus* species (*Quercus* sp.). The models demonstrated that annual variability explains a larger proportion of the seed count variability than spatial variability among plots. This is evidenced by the R-squared values, where annual factors alone explained 84% and 72% of the variability in seed counts in beech and oak forest areas, respectively. When both annual

and spatial factors were considered, the variability explained increased only to 86% for beech forest areas and 91% for oak forest areas. As expected, these findings suggest that the seed availability of both tree genera varied significantly among the years studied.

The seed availability patterns over time were similar within the plots of each tree species. Thus, a certain interannual synchrony in the availability of seeds was observed (Figure 3). During the study period, *Quercus* seeds were observed each year (mean = 500.73 seeds/5 m^2 ± 435.77 SD, C.V. = 87%), but *Fagus* seeds were only observed in five years (mean = 236.26 seeds/5 m^2 ± 364.01 SD, C.V. = 154%) and showed higher heterogeneity in their availability.

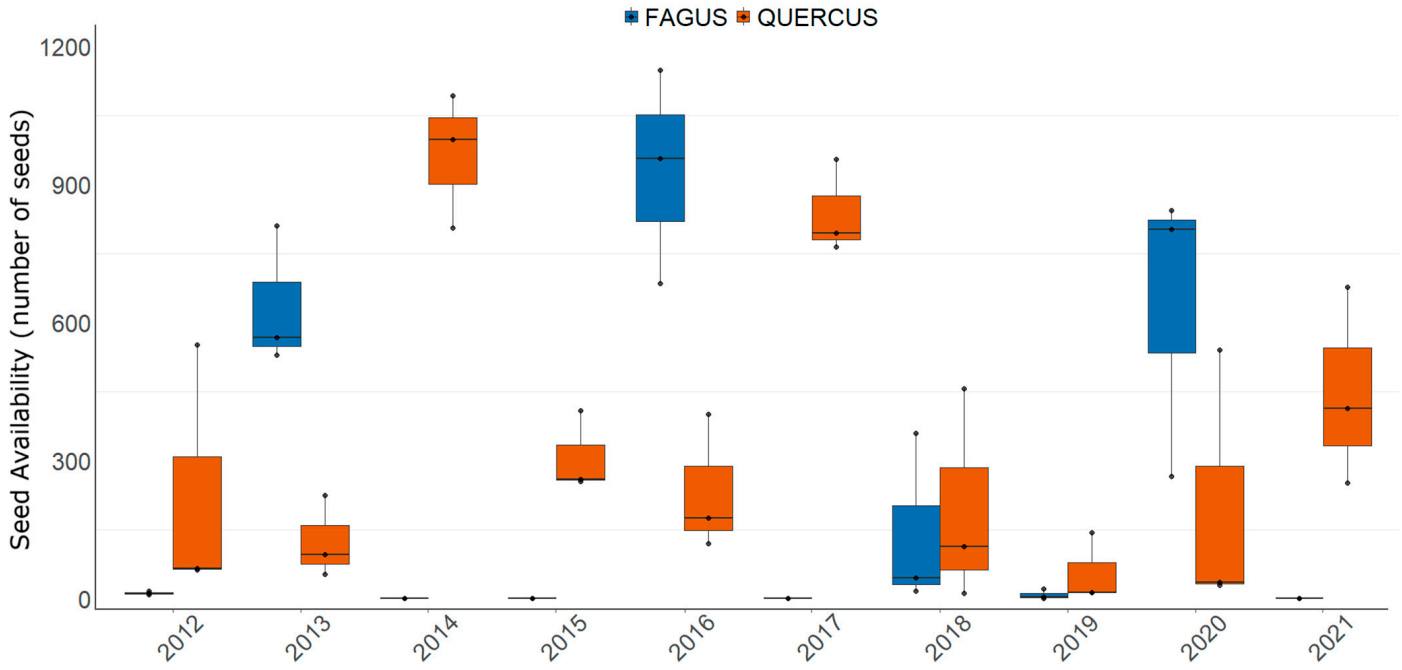


Figure 3. Seed availability per year for the main tree species. The median number of seeds (±quartiles) of the two main seed-producing tree species counted over 10 years (2012–2021) in the six study plots (black dots indicate the value of seeds counted in each plot).

3.2. Small Mammal Captures

During 2012-2022, a total of 1007 small mammals were trapped (58% *A. sylvaticus*, 22% *C. glareolus*, 13% *A. flavicollis*, and 8% *C. russula*).

We obtained between one and three significant statistical models (GLMM and LMM) for each response variable (Appendix A), and the most parsimonious models are shown in Table 2.

The predictive performance of our models exhibited relatively low explanatory power with respect to the predictors, explaining only between 5% and 18% of the variance in the response variable. The random factors had a greater influence.

The abundance of *A. sylvaticus* was higher in the spring (Table 2). The interaction between seed availability of the previous year and season was significant in all selected models of this species (Appendix A), suggesting a late response to seed dropping during the following spring. Seeds from the same year also had significantly positive effects, indicating an instant response to seed availability.

Table 2. Statistical models (GLMM and LMM) selected for each small mammal species and species richness (response variables). Predictors: season (autumn as reference), seed availability of the current year, seed availability of the previous year. Coefficients represent the effects of predictor variables on species abundance (±SE). Significant coefficients are indicated as follows: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Variables	*A. sylvaticus*	*A. flavicollis*	*C. glareolus*	*C. russula*	Species Richness
					
(Intercept)	0.92 ***	−1.48	−1.89 ***	−0.43 ***	1.91 ***
	(0.01)	(0.88)	(0.01)	(0.01)	(0.44)
SeasonSpring	0.82 ***	1.08 ***			0.39 *
	(0.01)	(0.00)			(0.15)
Seed of the same year	0.29 ***	0.69 ***	−0.68 ***		
	(0.01)	(0.00)	(0.01)		
Seed of the previous year	−0.03 *	0.27 ***	0.66 ***	0.44 ***	0.25 **
	(0.01)	(0.00)	(0.01)	(0.01)	(0.09)
Spring: seed same year		−0.70 ***			
		(0.00)			
Spring: seed previous year	0.59 ***	0.51 ***			
	(0.01)	(0.00)			
AICc	632.41	275.99	286.70	308.97	198.71
Dispersion ratio	0.88	1.11	0.66	0.76	-
Num. Obs.	114	114	114	114	80
Family	Negative binomial	Negative binomial	Negative binomial	Negative binomial	Gaussian
R^2 marginal	0.17	0.18	0.11	0.05	0.09
R^2 conditional	0.62	0.83	0.91	0.47	0.58

Regarding seed availability, we observed that seeds of the same and the previous year had significant effects on the abundance of *A. flavicollis* (Table 2). The response to seed availability in the same year was positive in the autumn, while it was positive in the spring after the seed fall in the previous year. This showed that in the year of seed fall, there was a population response in the autumn that extended until the next spring, but there was no anticipated response.

Both seed availability in the year of seed fall and the year after seed fall affected the abundance of *C. glareolus*, with the former having a negative effect and the latter having a positive effect (Table 2). Despite strong interannual variation in population abundance (Figure 4), this species did not show clear seasonal population changes.

The abundance of *C. russula* showed a positive response to seed availability in the previous year (Table 2).

Species richness (Order 0) increased in the spring and was associated with the seeds available in the previous year (Table 2). However, season and seed availability did not have effects on community diversity (Order 1) or evenness (Order 2); thus, no model appears in Table 2 for these two response variables.

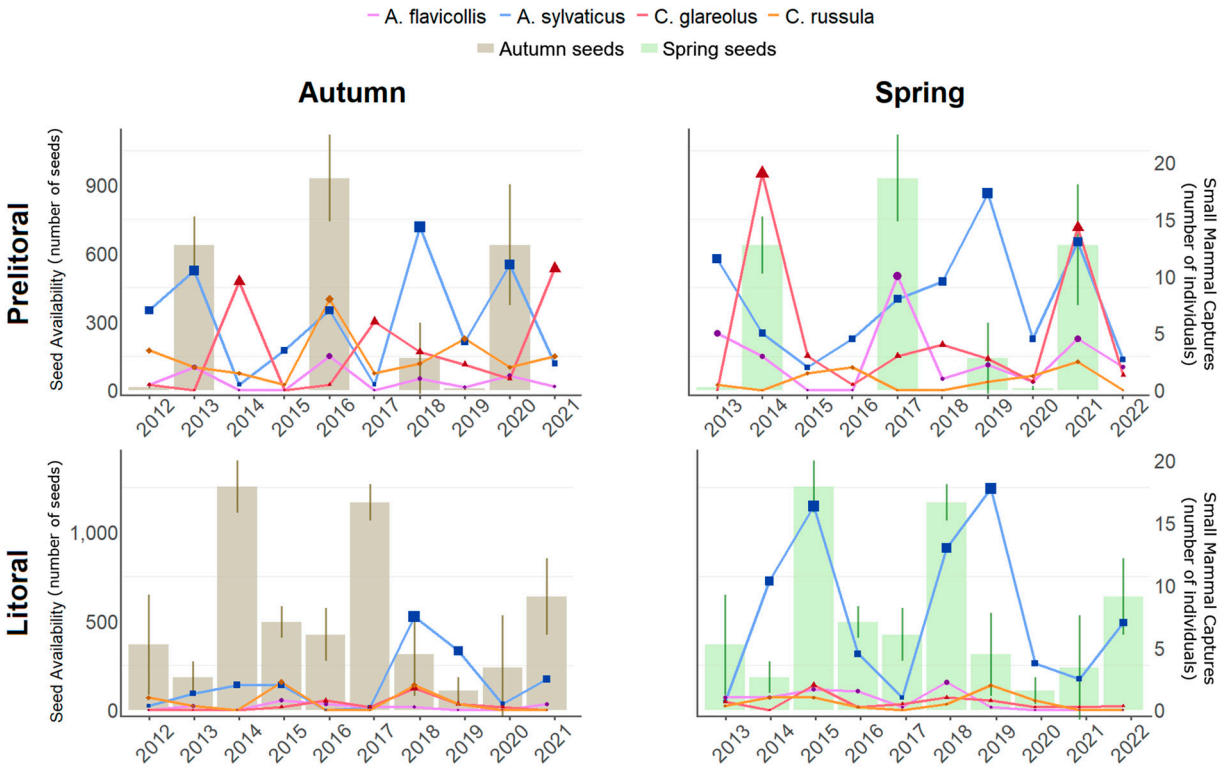


Figure 4. Mean number of seeds (bars ± SE) available within the same year (in autumn; brown colour) and in the previous year (in spring; green colour) and average number of captures (lines) of the four small mammal species in the spring and autumn.

4. Discussion

Our seed-producing species, *F. sylvatica* and *Quercus* sp., exhibited fluctuating seed availability, varying among years, being very high in some years and scarce in others. Previous studies have identified several climatic variables that influence seed production in these species. In the case of *F. sylvatica*, high seed production has been linked to summer temperatures in the two years prior to the production year [46,48,49]. In contrast, other climatic variables associated with variable seed production in *Quercus* sp. have been identified, such as high spring temperatures that synchronise flowering or annual differences in summer drought [26,97].

The results showed that variability in seed availability had important implications for the population dynamics of small mammals that depend (directly or indirectly) on these food sources, specifically, *A. sylvaticus*, *A. flavicollis*, *C. glareolus*, and *C. russula*, the four main species of small mammal communities found in forests of the study area [53]. These species revealed distinct patterns in responding to seasonality and variability in seed availability of the most frequent seed producers. Our results suggested that life history traits, feeding strategies, and foraging behaviour, were involved in the particular population responses of small mammal species to seed availability in Mediterranean mixed forests.

A. sylvaticus was the most abundant species in the communities studied, as was previously found in the area [53,55]. As a seed predator, this species has a relevant role in forest regeneration owing to its seed disperser activity [98]. The wood mouse is a generalist species and exhibits considerable plasticity in terms of feeding, using the most abundant source available [37,99,100]. This species showed a positive response to seeds within the same year, showing population responses to food availability and suggesting arboreal behaviour [62], as well as changing their breeding effort based on the availability of resources at the moment [101]. Studies performed in Mediterranean mixed forests revealed a strong association between seed availability and wood mouse dynamics during the autumn–winter period, with populations increasing towards the following spring [37,46].

Our results agreed with former studies [37,46] that also revealed a strong positive response to seeds of the previous year, which affected the abundance in the following spring. This pattern could be associated with seed scatter-hoarding behaviour, which allows the storage of acorns during the seed-falling season (autumn), and probably increases winter survival after feeding on seed caches well outside the period of natural seed availability [102]. This response suggested that the species takes full advantage of food availability throughout the year, affecting the length of its reproductive season [37,55,99]. In the Mediterranean, this species shows reproductive activity from late autumn to spring but, depending on suitable weather conditions, it can be extended towards the summer [37,87]. *A.sylvaticus* displayed consistent population dynamics, with boom and bust dynamics [91], increasing from autumn to spring and decreasing from spring to autumn because of summer breeding latency due to the characteristics of the Mediterranean climate [73,87,103]. This plasticity in food choice could be an advantage in changing environments, where seed availability is variable [104].

Similar to *A. sylvaticus*, *A. flavicollis* may eventually feed on small invertebrates, but its main diet consists of high-energy seeds and fruits [33]. Therefore, yellow-necked mice responded to seed availability either in the autumn or in the spring of the following year. Interestingly, this was the only studied species showing a stronger population response to the availability of seeds in the autumn of the current year (i.e., mice abundance versus seed availability in autumn) than to the availability of seeds of the previous year. In fact, *A. flavicollis* seems to be more adapted to the extraction of seeds from beech and other deciduous trees before maturation and dropping, a behaviour closely related to its arboreal habits and ease of climbing [100,105,106]. This species has very strict forest requirements and prefers deciduous tree-covered habitats [107,108]. *A. flavicollis* can also extend its reproductive period based on food availability and weather, which are key factors in the biology of the species [105,106,109]. This immediate response may be related to a strategy of maximising energy to survive during colder months or when other resources are scarce, with seeds being the food in which it specialises [38,110]. Thus, the rapid population response of *A. flavicollis* suggests a strategy of taking advantage of peaks in seed production while also ensuring forest regeneration by immediate seed dispersion.

C. glareolus dynamics were affected by the availability of seeds from the current (negative) and previous (positive) years. This means that the lowest densities were detected during the years of seed availability and the highest densities in the years without seed availability but following a year of seed production. This apparent contradiction can be observed after modelling rodent populations under seed production events [102]. *C. glareolus* is a herbivorous forest specialist in the study area [111], primarily feeding on leaves and seeds, although it may consume insects and earthworms [35,102]. In temperate regions, tree species have years in which seed production is very high, and such a year is followed by several years with very low production, showing fluctuating cycles [33]. All previous studies recorded a response of this species in the year following seed fall [33–35,57] because years of high seed availability produced a clear change in the age structure and densities of bank voles in the following spring and autumn. Because of this intense delayed response, the populations of this species probably exhibit self-regulation mediated by density dependence, producing population outbreaks or crashes regardless of food supplies [35,82]. Therefore, the strong dependence on seed availability produced significant interannual variations in abundance, but without seasonality (i.e., similar abundance in the spring and autumn).

In contrast, *C. russula* is a fascinating example of how interactions between small mammals and other species in their environment can interrelate in the Mediterranean forests. Although traditionally classified as an insectivore, this shrew has demonstrated an interesting and previously poorly documented response to seed availability, highlighting the importance of understanding the complexities of species ecology in the context of fluctuating resource production. First, it is essential to consider the relationship between the biology of the white-toothed shrew and its response to variability in seed production,

because its diet in Europe consists primarily of soil invertebrates [64,110]. This means that the response of *C. russula* to seed availability could not be direct; however, an increase in seed production may result in higher numbers of seed-predatory insects in the autumn, both through increased reproduction of these species and aggregation in seed-rich trees [112–114]. Both *Quercus* and *Fagus* trees suffer from invertebrates specialised in seed predation, where *Curculio elephas* and *Cydia fagiglandana* are some of the most important [33,115,116]. This coincides precisely with the population peak of *C. russula*, suggesting a relationship between its demography and the increase in invertebrates due to seed production, increasing the recruitment of juveniles in the summer after a seed fall year [55,60,110]. A similar situation occurred with house mice (*Mus musculus*) in New Zealand [117]. Furthermore, shrew populations will benefit from the increase in infested acorns on the ground as far as the season progresses since rodents leave infested acorns in situ [46]. Although the results are suggestive, we need to stress that forests are unsuitable habitats for *C.russula*, which shows a strong preference for scrublands [55,60,118] and a spatial retraction of its distribution range in the coastal area because of afforestation [55,60]. This means that the results obtained need to be considered with caution and that responses to seed availability could be also related to masked effects of weather on insect dynamics.

Regarding the response of the whole community, species richness increased in spring, probably because of the increased abundance of both mice (wood mice and yellow-necked mice) in that period. Additionally, the positive response to the availability of seeds in the previous year could be due to the population densities of the species normally encountered, where wood mice are the dominant species, and the rest are underrepresented. Consequently, wood mice abundance and their variations probably play an important role in explaining the community richness results. On the other hand, considering the Order 1 and 2 diversity indices, which are influenced by species evenness and dominance, no explanatory variable was included. This suggested that the representation of species in the community structure was not influenced by the variables tested. Other factors may have played relevant roles in the study area.

Given the dependence of seed production on local weather conditions at our study sites, it is crucial to pay attention to these factors, especially considering climate change predictions that could affect the region. It has been projected that increasing temperatures and drought episodes will result in a pronounced shift towards drier conditions in the region [119]. An increase in temperature has already been observed in Montseny in recent years [120,121]. These climatic changes may have a significant impact on *Quercus* species, with possible consequences for fauna that depend on these resources and for upstream ecological processes in forest ecosystems [122]. Previous studies have evaluated the effects of drier conditions on Mediterranean oak species and found a decrease in acorn production and greater variability among years [26,123,124]. These changing conditions can directly or indirectly alter the dynamics of forest ecosystems. The response of small mammals to seed availability not only reflects their ecological importance but also shows the challenges faced by these species. In a scenario of reduced seed production due to adverse climatic factors, small mammal populations will experience difficulties in their survival and persistence. This interdependence underscores the pressing need to understand and monitor these interactions in the context of climate change to implement effective conservation strategies that will safeguard biodiversity and the functionality of forest ecosystems in the future.

5. Conclusions

This study highlights how seed production variability can significantly influence the ecology of small mammals. In the face of projections indicating contractions in the potential distribution of all species studied [125], ranging from moderate in the shrew to large in all three rodents, climate change has emerged as one of the greatest challenges to biodiversity and ecosystem functioning globally. With the Mediterranean region predicted to experience an increase in the intensity and frequency of extreme weather events, seed production will certainly be affected, triggering a cascade of effects in forest ecosystems involving

multiple trophic levels [126–128]. This scenario will directly affect food availability for small mammals, which play critical roles in the ecosystem [10,129], such as seed dispersal, nutrient transfer, and regulation of insect populations [125], as well as serving as prey for several local predators [53,88]. In this context, our study highlights the importance of considering species interactions, resource availability, and climate change in the conservation and management of Mediterranean forest ecosystems. With the increasing need to address the conservation and functioning of ever-changing ecosystems, understanding how seed production, small mammals, and climate interact is a fundamental step in developing effective management and conservation strategies. This research provides a foundation for future studies and the implementation of adaptive conservation measures to ensure the resilience of forest ecosystems to the challenges of climate change.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/f15071148/s1, Table S1. Raw data materia.

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Appendix A. Complete Species Model Tables

Table A1. Models and predictor variables for Apodemus sylvaticus. Numbers indicate the rank order of statistical support of the selected models. Predictors: season (autumn as reference), seed availability of the current year, seed availability of the previous year. Coefficients represent the effects of predictor variables on species abundance. Values in parentheses are standard errors. Significant coefficients are indicated as follows: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

	Model 1	**Model 2**	**Model 3**
(Intercept)	0.92 ***	1.03 *	0.92 ***
	(0.01)	(0.40)	(0.01)
SeasonSpring	0.82 ***	0.77 ***	0.84 ***
	(0.01)	(0.19)	(0.01)
Seed of the same year	0.29 ***		0.38 ***
	(0.01)		(0.01)
Seed of the previous year	−0.03 *	−0.14	−0.03 *
	(0.01)	(0.15)	(0.01)
Spring: seed same year			−0.18 ***
			(0.01)
Spring: seed previous year	0.59 ***	0.57 **	0.56 ***
	(0.01)	(0.18)	(0.01)

Table A1. *Cont.*

	Model 1	Model 2	Model 3
AICc	632.41	633.44	634.04
Delta	0.00	1.03	1.63
Dispersion ratio	0.88	0.87	0.91
Num. Obs.	114	114	114
R^2m	0.17	0.14	0.17
R^2c	0.62	0.53	0.62

Table A2. Models and predictor variables for Crocidura russula. Numbers indicate the rank order of statistical support of the selected models. Predictors: season (autumn as reference), seed availability of the previous year. Coefficients represent the effects of predictor variables on species abundance. Values in parentheses are standard errors. Significant associations are indicated as follows: *** $p < 0.001$; * $p < 0.05$.

	Model 1	Model 2	Model 3
(Intercept)	−0.43 ***	−0.24	−0.27
	(0.01)	(0.64)	(0.66)
SeasonSpring		−0.41	−0.39
		(0.32)	(0.33)
Seed of the previous year	0.44 ***	0.40 *	0.55 *
	(0.01)	(0.19)	(0.22)
Spring: seed previous year			−0.36
			(0.30)
AICc	308.97	309.63	310.47
Delta	0.00	0.66	1.50
Dispersion Ratio	0.76	0.77	0.75
Num. Obs.	114	114	114
R^2m	0.05	0.06	0.06
R^2c	0.47	0.46	0.48

Small mammal community diversity.

Table A3. Models with diversity indexes and predictor variables. Numbers indicate the rank order of statistical support of the selected models. Predictors: season (autumn as reference), seed availability of the previous year. The results of the diversity indices in relation to the predictor variables. Significant associations are indicated as follows: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

	Order 0 (Richness)			Order 1 (Shannon)		Order 2 (Simpson)
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 1
(Intercept)	1.91 ***	2.11 ***	1.90 ***	0.50 **	0.50 **	0.31 **
	(0.44)	(0.42)	(0.38)	(0.16)	(0.19)	(0.10)
SeasonSpring	0.39 *		0.37 *		0.10 *	
	(0.15)		(0.16)		(0.04)	
Seed of the previous year	0.25 **	0.24 **				
	(0.09)	(0.09)				
AICc	198.71	200.63	200.70	75.94	77.79	−3.34
Delta	0.00	1.92	1.99	0.00	1.85	0.00
Num. Obs.	80	80	80	80	80	80
R^2m	0.09	0.06	0.04	0.00	0.01	0.00
R^2c	0.58	0.54	0.50	0.46	0.49	0.48

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